Aerospace Systems And Ethics Midterm

Fall 2026

This is a take home exam. You may work on it until the due date/time. You may use the course textbook, course materials, and any notes you took in class. You may not use the Internet (other than the textbook), consult with anybody else, or consult any AI resource in any way. You may use a calculator of your choice or Excel strictly for doing numerical calculations.

On all problems state a specific answer, as a number or an equation, circled or boxed or otherwise highlighted. Explain your work to be considered for partial credit. When giving numerical answers provide four significant figures.

For the first three problems consider an aircraft, like an earlier generation business jet, with the following specification sheet:

Parameter	Value	Units
Empty Mass	4100	kg
Fuel Capacity	3280	kg
Payload Mass	700	kg
Wing area	24	m^2
Thrust Per engine	16	kN
Number of engines	2	
Cruising speed	730	km/h
Cruising altitude	12000	meters
L/D max	15	
Engine Isp	4000	seconds

1. What is the Breguet range of the aircraft with a 10% fuel reserve? Give a number.
2. Suppose we built a supersonic version capable of Mach 2 cruising (2100 km/hr). Assuming the mass numbers are the same but other numbers change to be consistent with limits on supersonic flight, what would the Breguet range be? Give a number and explain which parameters you changed and why.
3. Suppose we built a hypersonic version capable of Mach 5 cruising (5000 km/hr). Assuming the mass numbers are the same but other numbers change to be consistent with limits on hypersonic flight, what would the Breguet range be? Give a number and explain which parameters you changed and why.

For the next five problems assume the following for the aircraft above:

The lift coefficient is $C_L = L_\alpha \alpha$. In other words, the lift coefficient is a constant times the angle of attack. This applies for angles of attack over the range zero to α_{stall}.

The drag coefficient is $C_D = C_{D_0} + k * C_L^2$.

The numerical values of the parameters are:

Parameter	Value	Units
CD0	0.017	None
k	0.065	None
Lalpha	0.075	per degree
Stall AOA	20	deg
Air density at 12k meters	0.312	kg/m^3
Air density at 0 meters	1.225	kg/m^3

4. Using these numbers, what is the maximum L/D? Would you call that consistent with the specification sheet for the aircraft?
5. Assuming these numbers, and that the aircraft cruises at the max L/D point at its takeoff weight, what is the computed airspeed for the cruising condition at 12,000 meters? Would you call that consistent with the specification sheet for the aircraft?
6. Assuming these numbers, what is the thrust setting at cruise conditions?
7. What is the minimum steady, level flight speed of the aircraft at sea level? What is the thrust setting at the minimum flight speed?
8. At sea level can this airplane sustain a climb angle of 45 degrees? If so, with the engines running at full thrust, what is its rate of climb?

Suppose we have a quadcopter (four rotors) with weight W and a thrust of up to Tmax newtons per rotor (Tmax > W/4). To fly the quadcopter we tip it forward at an angle theta.

9. What is the relationship between the constant pitch angle of the quadcopter and forward velocity for steady level flight? Assume a constant drag coefficient and no lift coefficient. Define all terms in a diagram.
10. What is the maximum pitch angle that can be flown stably?

Suppose we mount a wing with area S on the quadcopter. The wing is mounted so when the pitch angle is zero (the quadcopter is level) the wing is at the stall angle relative to forward

flight. Now we fly the quadcopter by pitching it forward and adjusting the thrust of the rotors as needed.

11. What is the equilibrium steady, level flight condition as a function of pitch angle and thrust settings? State your assumptions about how the presence of the wings changes the lift and drag characteristics of the previously wingless quadcopter.
12. The quadcopter with the wing has more drag when flying than the wingless quadcopter, but it might need less thrust to stay aloft than the wingless. Assuming the same amount of thrust is available, is the winged quadcopter faster or slower than the wingless? If the answer is ambiguous what does it depend on?